

DRY-RUN GUIDE | q1.py

Source: Python\3rd_Belt\Matrix\q1.py

CORE BEHAVIOR

Adds corresponding cells of same-sized matrices; raises ValueError when dimensions differ.

REPRESENTATIVE INPUT

[[1,2],[3,4]] and [[5,6],[7,8]].

STEP TRACE

Compute cells: 1+5=6, 2+6=8, 3+7=10, 4+8=12.

EXPECTED OUTPUT / RETURN

Rows '6 8' and '10 12'.

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\3rd_Belt\Matrix\q1.py; line numbers refer to the original file. No source was modified.

```
001 | # Question: Add two matrices of equal size.
002 | # Add your solution here.
003 |
004 |
005 | def add_matrices(matrix1, matrix2):
006 |     if len(matrix1) != len(matrix2) or len(matrix1[0]) != len(matrix2[0]):
007 |         raise ValueError("Matrices must have the same dimensions")
008 |
009 |     rows = len(matrix1)
010 |     cols = len(matrix1[0])
011 |     result = [[0 for _ in range(cols)] for _ in range(rows)]
012 |
013 |     for i in range(rows):
014 |         for j in range(cols):
015 |             result[i][j] = matrix1[i][j] + matrix2[i][j]
016 |
017 |     return result
018 |
019 |
020 | if __name__ == "__main__":
021 |     rows, cols = map(int, input().split())
022 |     matrix1 = [list(map(int, input().split())) for _ in range(rows)]
023 |
024 |     rows2, cols2 = map(int, input().split())
025 |     matrix2 = [list(map(int, input().split())) for _ in range(rows2)]
026 |
027 |     result = add_matrices(matrix1, matrix2)
028 |     for row in result:
029 |         print(*row)
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.